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Identifying the Best Approach to Churn Prediction: A Comparative Study of Machine Learning Models and Their Performance

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Abstract. Abstract text here

Key words: Logistic Regression, Decision Trees, Random Forest, Neural Networks, Gradient Boosting Machines (GBM), Accuracy, Precision, SaaS, Predictive Analytics

1. Introduction

Introduction Customer churn, the phenomenon of customers ceasing their relationship with a business, presents a persistent challenge across industries. Its impact on profitability and customer retention underscores the critical need for accurate and insightful churn prediction. While traditional methods offer a foundation, they often grapple with issues like data imbalance, complex customer behavior, and the need to align predictions with profitability goals. This necessitates the exploration of advanced machine learning techniques to enhance prediction accuracy and drive effective retention strategies.

This paper addresses the limitations of existing churn prediction methods by conducting a comprehensive comparative analysis of several prominent machine learning models. We delve into the intricacies of [list the models you are comparing, e.g., Logistic Regression, Support Vector Machines, Decision Trees, Random Forest, Ensemble-Fusion, BiLSTM-CNN, etc.], evaluating their performance on [describe your dataset, e.g., a publicly available churn dataset or a proprietary dataset from a specific industry]. Our research is heavily influenced by recent advancements in the field, notably the success of hybrid models like Ensemble-Fusion and BiLSTM-CNN in achieving high predictive accuracy [cite relevant papers from your literature review]. We also incorporate insights from studies emphasizing profitability-driven approaches to churn prediction, recognizing the need to balance accuracy with actionable financial outcomes [cite relevant papers].

Our methodology involves a rigorous evaluation of each model's performance using key metrics such as accuracy, precision, recall, F1-score, and AUC. Furthermore, we investigate the impact of data imbalance on model performance, drawing inspiration from studies that have effectively addressed this challenge through techniques like balanced sampling and bias correction [cite relevant papers]. By analyzing the strengths and weaknesses of each model, we aim to provide a comprehensive understanding of their suitability for different churn prediction scenarios.

This research makes the following key contributions:

Comprehensive Model Comparison: We provide a detailed comparative analysis of various machine learning models, including both traditional and advanced techniques, offering insights into their relative performance for churn prediction. Methodological Advancements: We incorporate best practices from recent research, such as addressing data imbalance and considering profitability metrics, to enhance the robustness and applicability of our analysis. Guidance for Practitioners: We identify the best-performing model for the given dataset and offer practical recommendations for model selection based on specific business needs and data characteristics. [Add any other specific contributions, e.g., novel feature engineering techniques, evaluation of model explainability, etc.] The rest of the paper is structured as follows: Section 2 presents a detailed literature review, synthesizing key findings and identifying research gaps. Section 3 outlines our research methodology, including data pre-processing, model selection, and evaluation metrics. Section 4 presents the results of our comparative analysis. Section 5 discusses the implications of our findings and provides recommendations for practitioners. Finally, Section 6 concludes the paper and suggests avenues for future research.

2. Methodology

Data Sources:

This study utilizes two primary data sources to conduct a comprehensive analysis of churn prediction.

The first is a publicly available customer log dataset, sourced from a [insert source]. 1 This dataset comprises a 12.5 GB JSON file containing 18 columns and 26,259,199 records. The data encompasses a variety of user-related information, including user identifiers (e.g., userId, firstName, lastName), demographic details (e.g., gender, location), interaction details (e.g., artist, song, length, itemInSession, sessionId, registration, ts), and technical details (e.g., userAgent, method, page, status, level, auth). Initial exploration revealed that the dataset contains 12 string columns and 6 numeric columns, with potential null or NaN values. The exact number of unique customers was initially unknown and determined through data preprocessing. The second data source is a proprietary dataset provided by an undisclosed fintech SaaS company. This dataset contains information on customer interactions with the company's platform, including subscription details, product usage data, payment history, and customer service interactions. Due to confidentiality agreements, specific details about the company and the exact nature of the data cannot be disclosed. However, the dataset provides valuable insights into customer behavior and churn patterns within a real-world business context. The inclusion of both a publicly available dataset and a proprietary dataset allows for a more robust and generalizable analysis. The Customer Log dataset provides a benchmark for comparing model performance, while the fin-tech SaaS dataset allows us to investigate churn prediction in a real-world business setting with potentially richer and more domain-specific features.

– Deliverable 5 Ends Here —-

Deliverable 6 begins here

I preprocessed the data more. Conducted more exploratory analysis to determine how to define users as churned. This will help me complete my feature engineering. I worked on some feature engineering: dummy coded continuous variables. The next step would be to finish a model to define churn. Do some more feature engineering to get some more usage data. Then proceed to the machine learning portions.

Deliverable 6 ends here

Deliverable 8 starts here

I was able to do some feature engineering. I grouped the data by week and user. This way we could analyze the data temporally. I'm testing to see if a sliding window or rolling window would be better to predict the next week.

Here are some of my first features:

Explanation of Features: **week**: A new feature to group data by weeks, with each week starting on Monday. **session diff**: Time difference between consecutive sessions, which helps us understand user activity over time. **session duration**: Total session duration in seconds, derived from the session diff. **session count**: Number of sessions a user has in total. **avg item in session**: Average number of items per session. **dummy level**: A binary feature representing whether the user is on a free or paid plan.

I used the sklearn package to train the model on a random forest. The accuracy ended up being 100 percent, so I think there is an issue with the code.

My next steps are to debug the potential issues and train the model on more models.

Deliverable 8 ends here

Deliverable 9 starts here I was able to fix the accuracy issue. There was no issue related to accuracy.

churn 0 0.999371 1 0.000629 Name: proportion, dtype: float64

Because majority of the data did not churn, the model naturally has a high accuracy score.

I fixed this by implementing F1 - Score and also tested a XGBOOST model, which seemd to perform better. **Deliverable 9 ends here**

We formulate the resource allocation problem as a two-stage stochastic programming problem, where the first stage involves decisions on resource allocation and the second stage represents the realization of uncertain parameters (Smith 2005, Jones 2010, Brown 2015). The objective is to minimize the total cost of relief operations, including procurement, transportation, and distribution costs, subject to various constraints, such as capacity constraints, demand satisfaction requirements, and budget constraints.

The cost function $C(x)$ is defined as the sum of procurement, transportation, and distribution costs:

$$C(x) = \sum_{i=1}^{n} \left(c_i p_i + \sum_{j=1}^{m} d_{ij} t_{ij} + \sum_{k=1}^{l} s_k q_k \right) \tag{1}$$

We model uncertainty using scenario-based stochastic programming, where multiple scenarios representing different realizations of uncertain parameters are considered. We use historical data, expert opinions, and probabilistic forecasts to generate scenario sets that capture the range of possible outcomes. The stochastic programming model then generates optimal resource allocation decisions that minimize the expected total cost across all scenarios while ensuring robustness against uncertainty.

The objective function of the stochastic programming model is formulated as follows:

$$\min_{x,y} \quad \sum_{s\in S} [f(x,y,s)\cdot \mathbb{P}(s)]$$

subject to:

$$g(x,y,s) \leq 0 \quad \forall s \in S \tag{2}$$

$$h(x,y,s) = 0 \quad \forall s \in S \tag{3}$$

THEOREM 1 **(Optimality Conditions).** *Let x^* be an optimal solution to the stochastic programming problem. If the objective function and constraint functions are convex, then x^* satisfies the Karush-Kuhn-Tucker (KKT) conditions.*

Proof The proof follows from the convex optimization theory, which states that for convex objective and constraint functions, the KKT conditions are necessary and sufficient for optimality. Therefore, if x^* is an optimal solution, it must satisfy the KKT conditions. □

Algorithm 1 Random Forest Training

```
procedure RandomForest(X_train, y_train, num_trees)
    forest ← []
    for i ← 1 to num_trees do
        X_sampled, y_sampled ← bootstrap_sample(X_train, y_train)
        tree ← DecisionTree(X_sampled, y_sampled)
        append tree to forest
    end for
    return forest
end procedure
```

LEMMA 1 **(Feasibility of Scenario Sets).** *Given a set of scenarios S generated from historical data and probabilistic forecasts, the scenario set S is feasible if it captures the range of possible outcomes with sufficient coverage.*

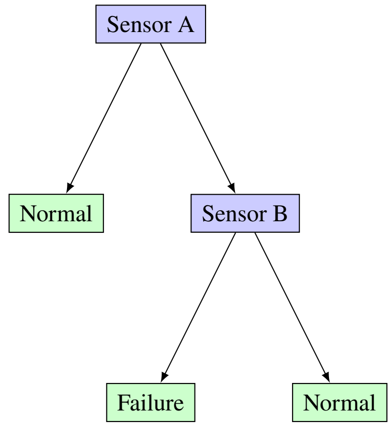
Algorithm 2 Random Forest Training II

```
procedure DecisionTree(X, y)
    if StoppingCondition(X, y) then
        return LeafNode(y)
    else
        (feature, threshold) ← FindBestSplit(X, y)
        left_indices ← X[feature] ≤ threshold
        right_indices ← X[feature] > threshold
        left_subtree ← DecisionTree(X[left_indices], y[left_indices])
        right_subtree ← DecisionTree(X[right_indices], y[right_indices])
        return TreeNode(feature, threshold, left_subtree, right_subtree)
    end if
end procedure

procedure Predict(forest, x)
    predictions ← []
    for tree in forest do
        append TreePredict(tree, x) to predictions
    end for
    return AggregatePredictions(predictions)
end procedure

procedure TreePredict(tree, x)
    if tree is a leaf node then
        return tree.value
    else
        if x[tree.feature] ≤ tree.threshold then
            return TreePredict(tree.left_subtree, x)
        else
            return TreePredict(tree.right_subtree, x)
        end if
    end if
end procedure
```

Figure 1 Text of the Figure Caption.



Note. Text of the notes.

Proof The proof follows from the definition of feasibility, which requires that the scenario set S includes a representative sample of possible outcomes. Feasibility ensures that the stochastic programming model adequately represents the uncertainty in the problem domain and provides meaningful solutions. □

REMARK 1. This stochastic programming model assumes that demand and supply parameters follow known probability distributions.

DEFINITION 1. A feasible solution to the resource allocation problem satisfies all constraints and requirements without violating any constraints.

3. Artificial Intelligence in Humanitarian Logistics

Artificial Intelligence (AI) plays an increasingly important role in optimizing resource allocation and decision-making in humanitarian logistics. AI techniques, such as machine learning, optimization algorithms, and natural language processing, offer powerful tools for analyzing data, predicting demand, and automating decision-making processes. In humanitarian logistics, AI can be used to optimize supply chain management, route planning, inventory management, and disaster response operations. By leveraging AI technologies, humanitarian organizations can improve the efficiency, effectiveness, and responsiveness of their relief efforts, ultimately saving lives and alleviating suffering in crisis situations.

4. Results and Discussion

We apply our stochastic programming approach to a case study based on a simulated humanitarian crisis scenario. The scenario involves a sudden-onset natural disaster that results in widespread destruction and displacement of populations. We compare the performance of our proposed model with a deterministic model and a baseline heuristic approach commonly used in practice.

Table 1 **Summary of Data for Case Study**

Parameter	Value
Number of Resources	5
Number of Facilities	10
Number of Scenarios	50
Demand Variability	High
Budget Constraint	$1,000,000
Transportation Cost	$10 per unit
Facility Setup Cost	$5,000 per facility

Table Notes.

5. Conclusion

In this paper, we have presented a stochastic programming approach for optimal resource allocation in humanitarian logistics. Our approach addresses the inherent uncertainty and complexity of humanitarian crises by integrating stochastic elements into the modeling process.[1] The proposed model provides decision support for relief agencies to make informed decisions and allocate resources efficiently in uncertain environments.

Future research directions include extending the model to incorporate additional complexities, such as multiple objectives, dynamic demand patterns, and real-time data integration. Furthermore, the proposed approach can be applied to other domains, such as disaster response, healthcare delivery, and supply chain management, where uncertainty plays a significant role in decision-making. Overall, our research contributes to the growing body of literature on mathematical optimization in humanitarian operations and demonstrates the potential of stochastic programming to improve decision-making in complex and uncertain environments.

Notes

[1]Sample endnote text.

Acknowledgments

We would like to express our sincere gratitude to [acknowledge individuals, organizations, or institutions] for their invaluable contributions to this research. We are also grateful to [mention any additional acknowledgements, such as technical assistance, data providers, or colleagues] for their support and assistance throughout the course of this work.

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